

EXPT NO:3	CLEANSE DATA BY HANDLING MISSING VALUES, OUTLIERS, AND INCONSISTENCIES USING TABLEAU AND POWERBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

- 1. What are the potential impacts of missing values in a dataset, and what are some common strategies to address them in Tableau and Power BI?**

Missing values can reduce the accuracy of reports and may lead to incorrect business decisions if they are ignored. They often create gaps in charts, calculations, and performance metrics. Before analyzing data, it is important to identify where values are missing. In Tableau and Power BI, missing values can be handled by replacing them with average or default values, filtering incomplete records, or removing unnecessary rows. Both tools also provide data preparation features to detect blank or null values easily. Proper handling of missing data improves the reliability of Business Intelligence dashboards and reports.

- 2. What is the importance of identifying and addressing outliers in a dataset? How can outliers affect data analysis and model performance?**

Outliers are data values that are much higher or lower than the rest of the dataset. These unusual values can distort averages, trends, and visualizations, leading to misleading conclusions. In predictive models, outliers may reduce accuracy and produce unreliable results. Identifying them helps analysts determine whether they are valid observations or data entry errors. Tableau and Power BI make it easier to detect outliers using charts such as box plots and scatter plots. Managing outliers ensures that business insights are more accurate and meaningful.

- 3. What types of data inconsistencies might you encounter during data cleansing, and how can Tableau and Power BI help in resolving them?**

During data cleansing, inconsistencies such as duplicate records, spelling mistakes, incorrect date formats, blank fields, and inconsistent text values are commonly found. These issues can reduce the quality of analysis and affect report accuracy. Tableau and Power BI provide tools to clean and transform data before creating visualizations. Features like Power Query in Power BI and Tableau Prep allow users to remove duplicates, standardize formats, and correct errors efficiently. They also help merge and organize data from multiple sources. Clean and consistent data leads to more reliable Business Intelligence outcomes.

- 4. How do you identify patterns in data using visualizations to spot missing values, outliers, and inconsistencies?**

Visualizations make it easier to understand data by presenting information in a clear and interactive way. Charts such as histograms, box plots, scatter plots, and heat maps can quickly highlight unusual patterns. Missing values may appear as blank spaces or gaps, while outliers are shown as points far from the main data distribution. Inconsistent values often become visible through unexpected categories or irregular trends. Tableau and Power BI allow users to apply filters and drill-down options to

investigate these issues further. This visual approach helps improve data quality before performing detailed analysis.

5. What is the role of calculated fields in data cleansing, and how can they be used to handle missing values and outliers in Tableau/Power BI?

Calculated fields are used to create new values based on existing data and are useful during the data cleansing process. They can replace missing values with meaningful defaults, averages, or custom calculations. They are also helpful in identifying outliers by setting conditions or creating threshold-based rules. In Tableau and Power BI, calculated fields allow users to automate these corrections without changing the original dataset. This makes the data more consistent and easier to analyze. As a result, reports and dashboards become more accurate, reliable, and suitable for business decision-making.

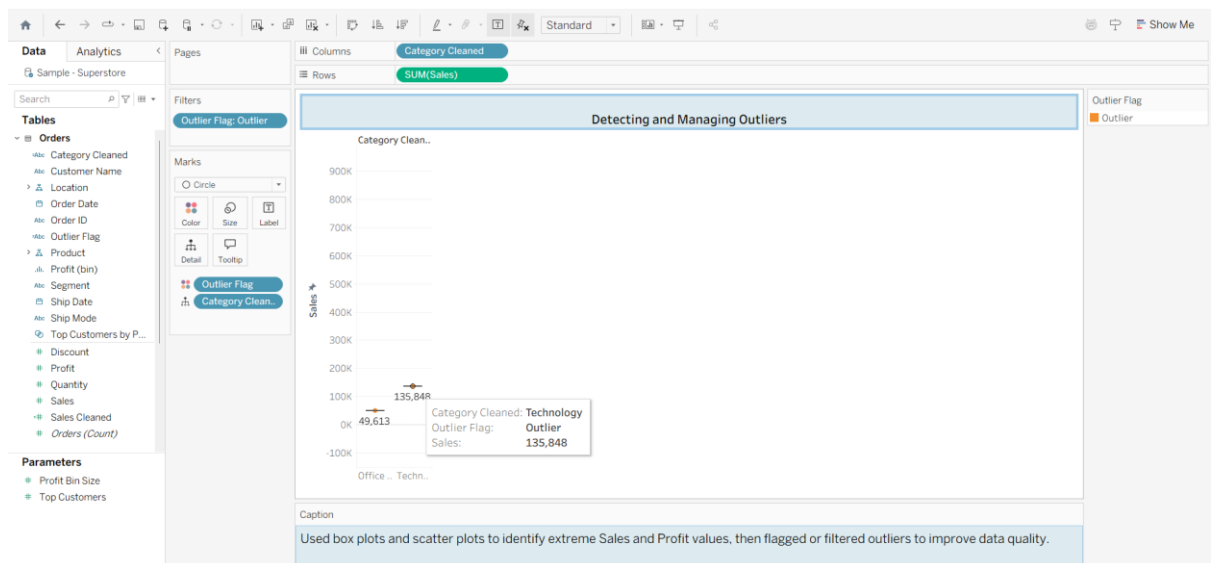
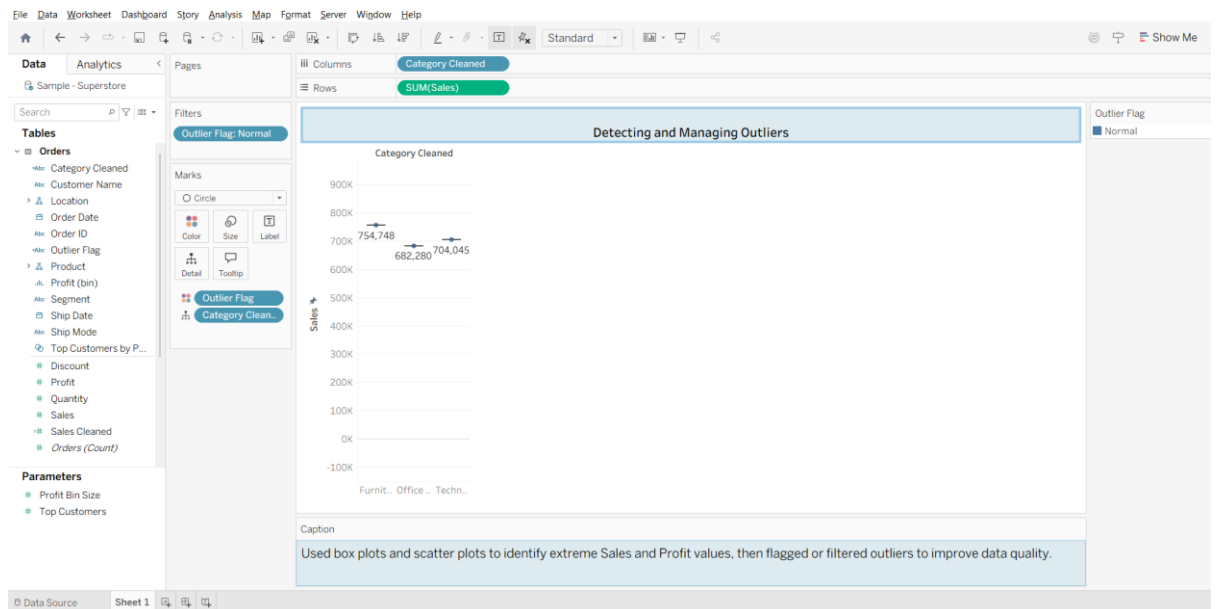
IN-LAB EXERCISE:

OBJECTIVE:

To clean a given dataset (e.g., Superstore Sample in Tableau and Financials/Sales Data in Power BI) by identifying and resolving missing values, outliers, and inconsistencies using built-in tools, ensuring the dataset is accurate and ready for analysis.

PROCEDURE:

1. **Open the Tool and Connect Dataset** ○ Open Tableau Desktop , Power BI Desktop.
 - Load the built-in dataset (Sample - Superstore in Tableau / Financials in Power BI).
 - Preview the dataset to understand its structure.
2. **Initial Inspection** ○ Explore fields such as Sales, Profit, Region, or Product Category.
 - Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.
3. **Handling Missing Values** ○ Use Data Interpreter (Tableau) / Power Query (Power BI) to highlight nulls. ○ Replace missing numerical values with mean/median, and text with "Unknown".
 - Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.



4. Detecting and Managing Outliers

- Create scatter or box plots for Sales and Profit to identify extreme values.

- Choose strategies:

- Filter them temporarily.
- Apply caps using calculated columns.
- Flag them with labels for review.

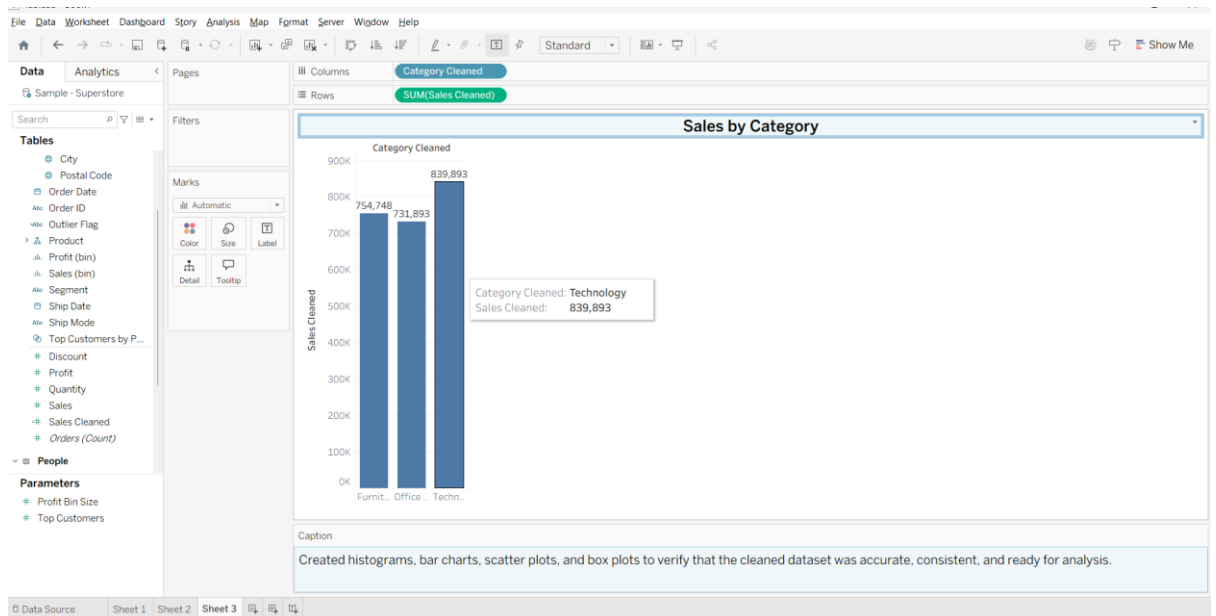
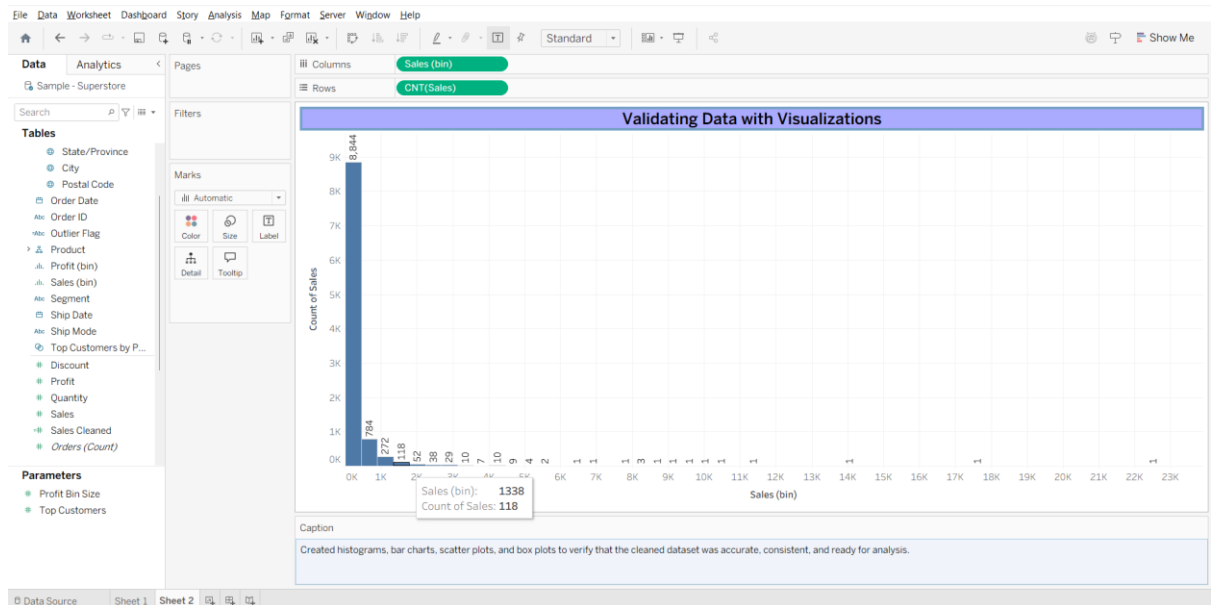
5. Resolving Inconsistencies

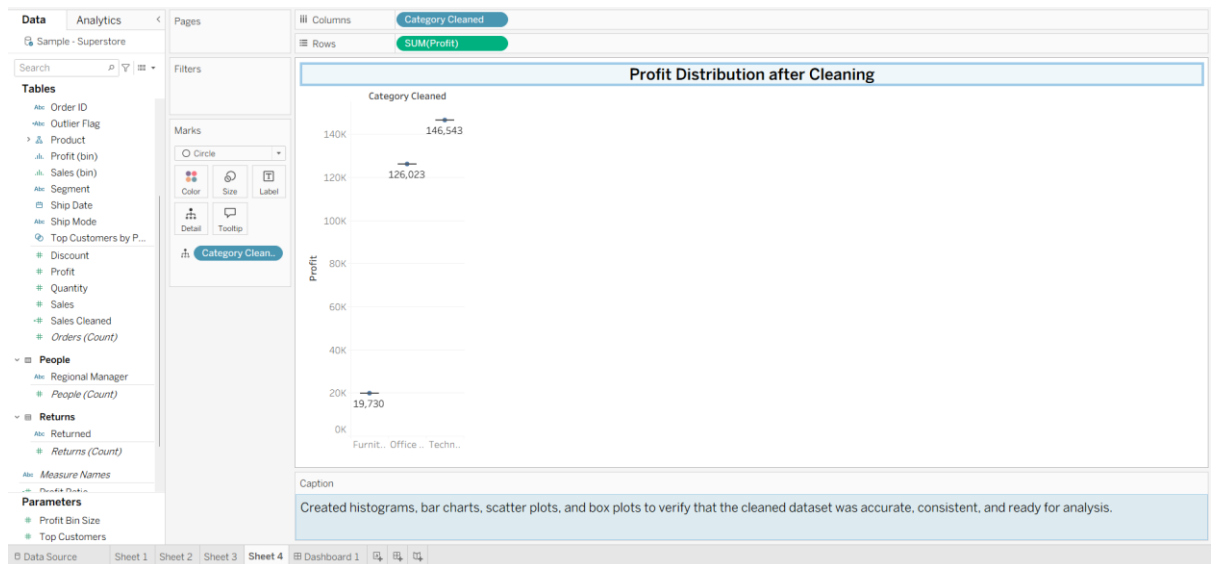
- Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.

- Normalize data formats, such as date fields or currency units.

6. Validation with Visuals

- Build dashboards with histograms, bar charts, and box plots to visually confirm improvements.
- Compare before/after cleaning stages for accuracy.





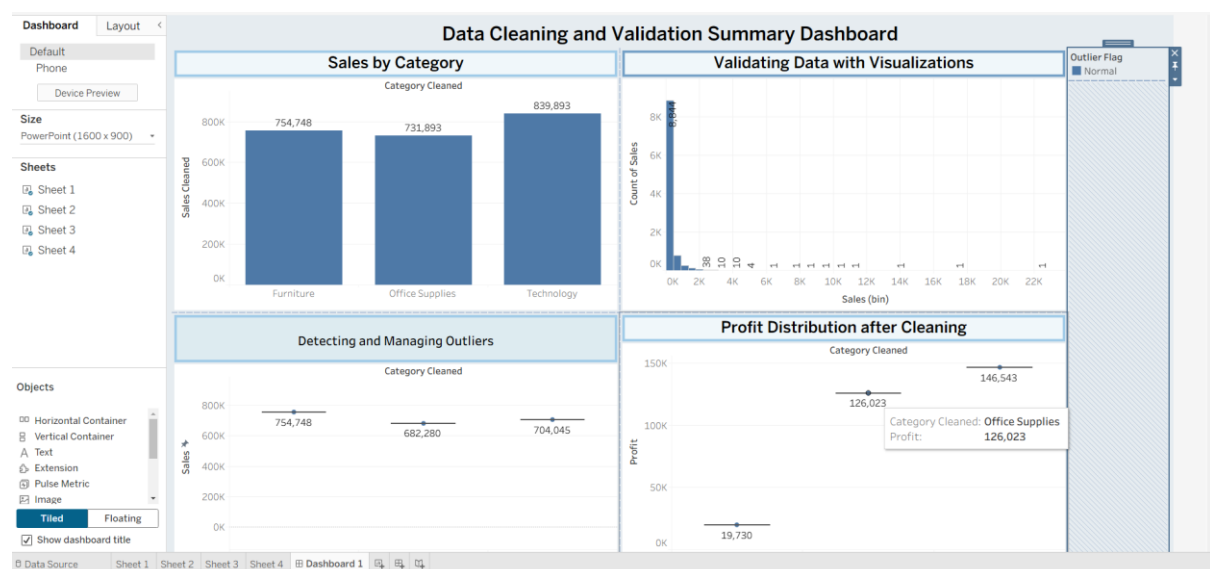
7. Save the Cleaned Dataset

Save as .twbx (Tableau) / .pbix (Power BI).

- Optionally export cleaned data to .csv.

8. Summary Dashboard

- Create a visual summary of the cleaning process, highlighting:
 - Fields corrected.
 - Nulls/outliers removed or adjusted.
 - Clean dataset metrics (record count, completeness %).



Experiment Dataset:

1. Tableau: *Sample - Superstore*
2. Power BI: *Financials or Sales Data*

How to Access:

Tableau:

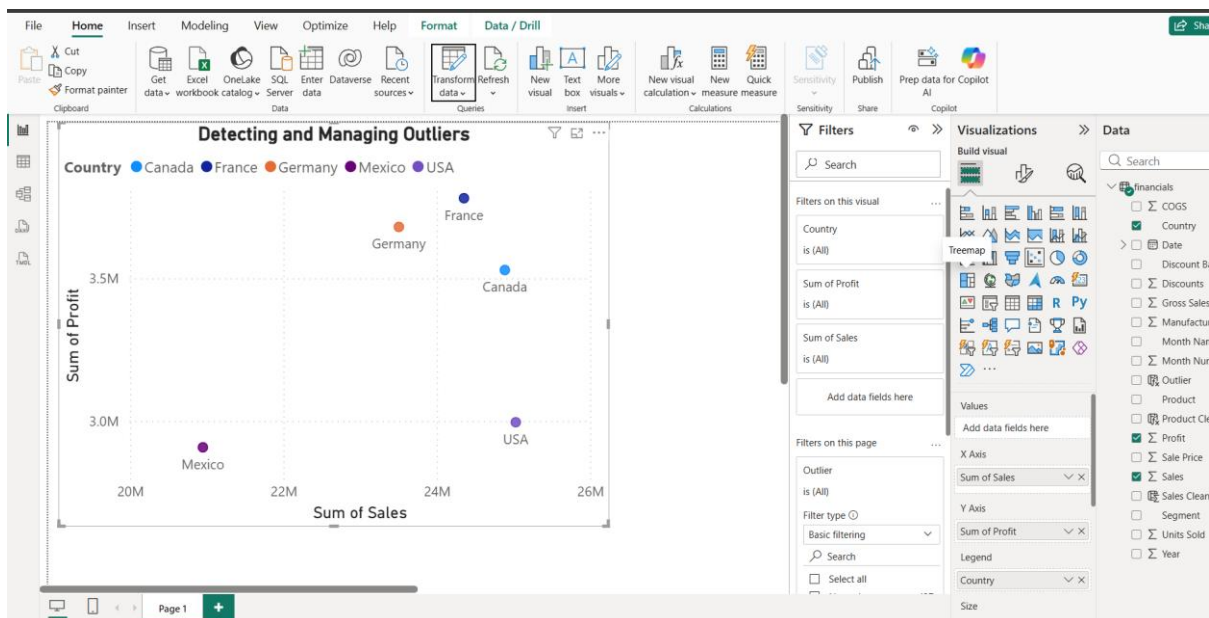
- **Navigate to:** *Start Page → Sample Workbooks*
- Or: *Help → Sample Data → Open Sample Data Folder*

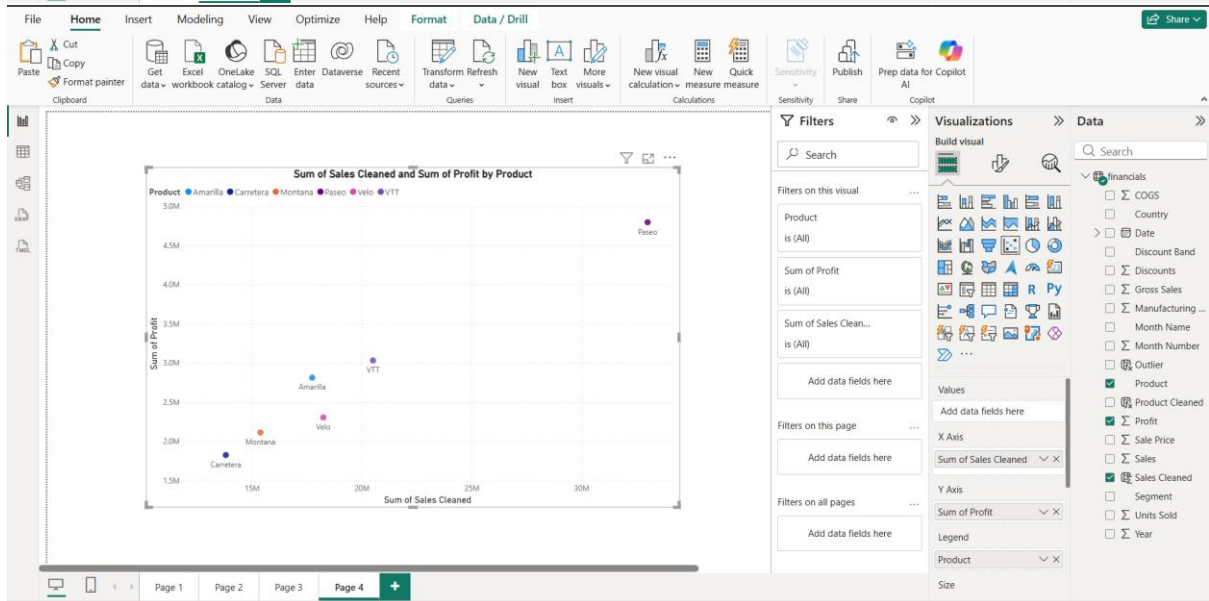
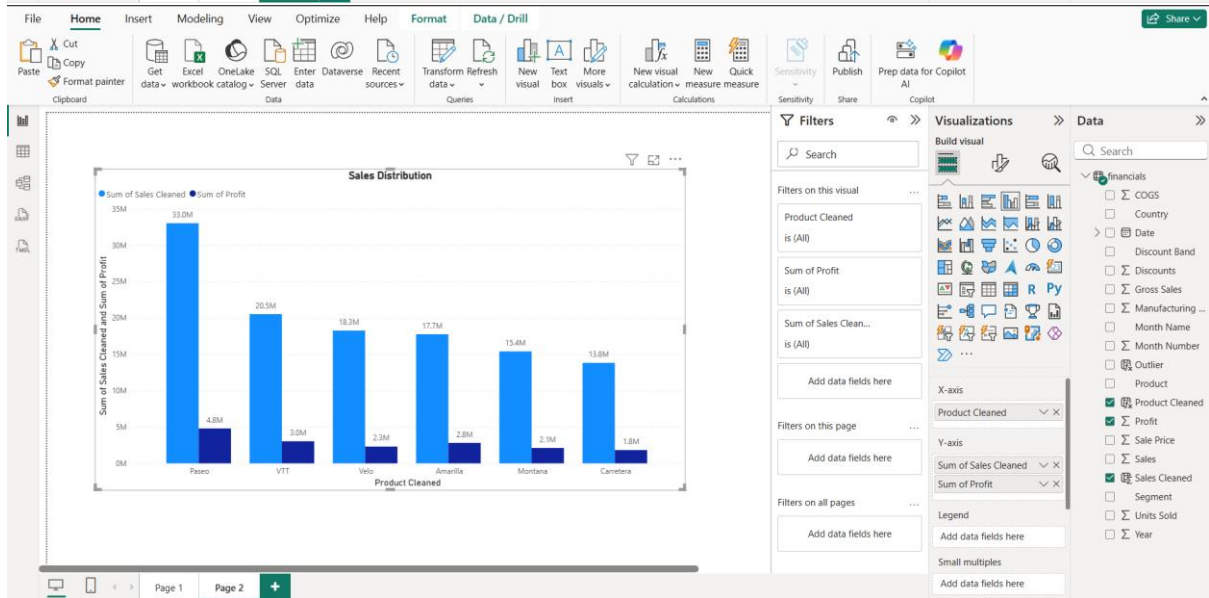
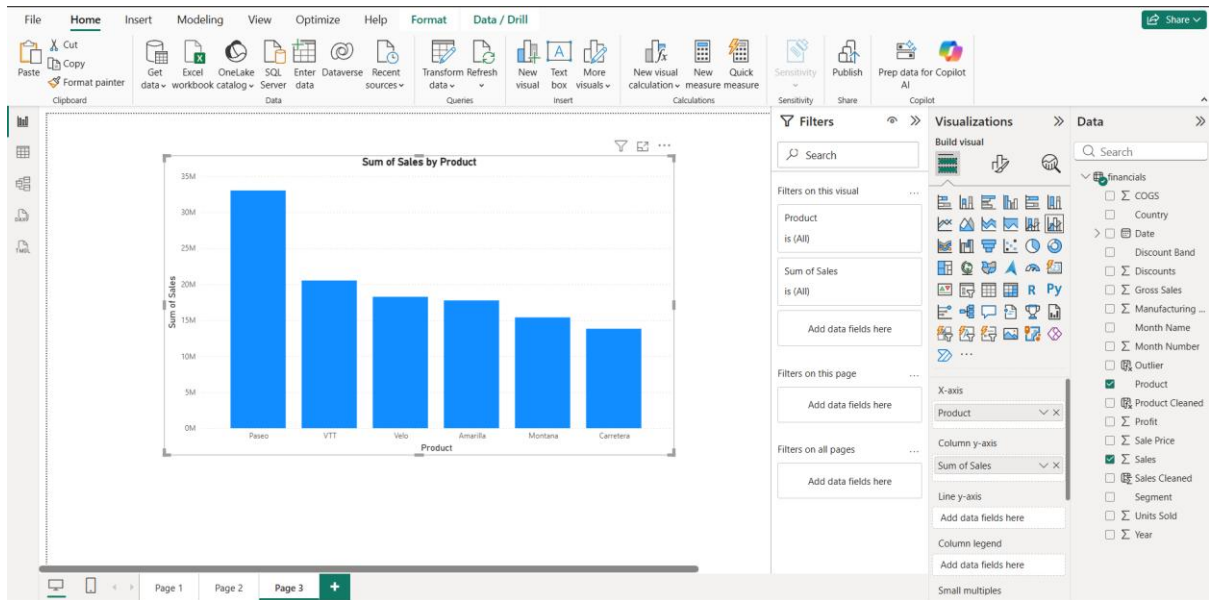
Power BI:

- **Navigate to:** *Home → Get Data → Excel → Choose sample datasets (e.g., Financial Sample.xlsx)*
- Microsoft also hosts Power BI sample datasets online.

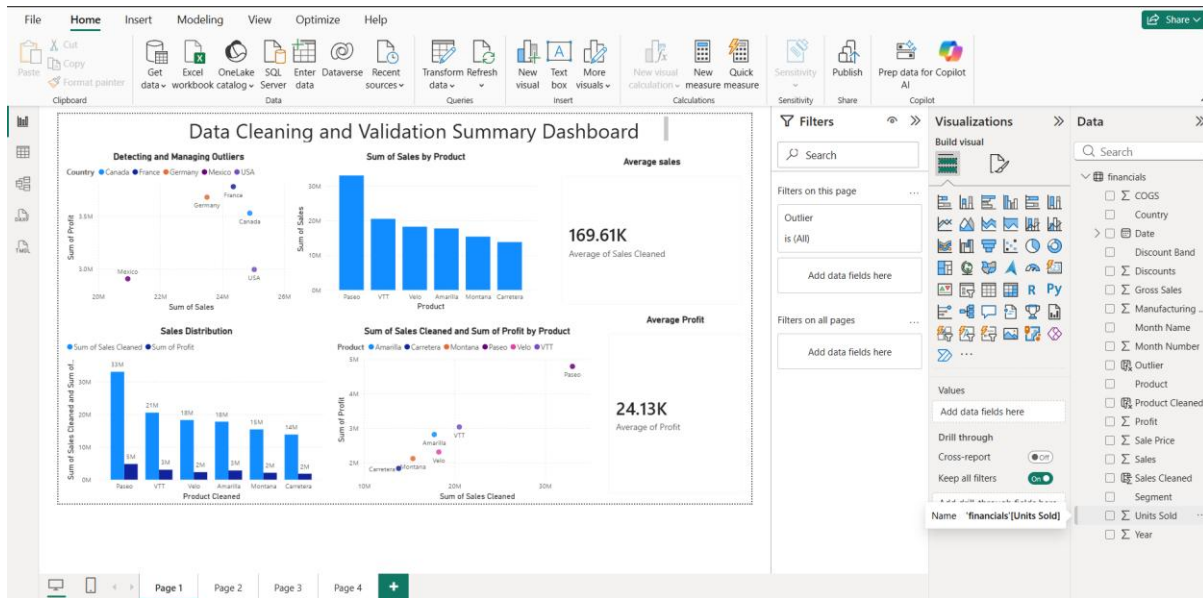
Fields Used:

1. Sales
2. Profit
3. Region
4. Category
5. Order Date / Transaction Date





Final Dashboard



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What method did you use to replace missing values in numeric fields?

In this experiment, missing values in numeric fields were handled by replacing them with suitable values instead of leaving them blank. The **mean (average)** value was used because it preserves the overall data distribution and avoids losing useful records. In Tableau and Power BI, null values were first identified and then replaced during the data preparation stage. This approach ensured that calculations and visualizations remained complete and accurate. By filling the missing values, the dataset became more reliable for analysis. As a result, the dashboard produced meaningful and consistent business insights.

2. How were outliers handled in your dataset and what visual confirmed their correction?

Outliers were identified by comparing values that were significantly different from the rest of the dataset. Instead of deleting these records, they were filtered or flagged so they could be analyzed separately without affecting the overall results. A **scatter plot** was used to detect unusual data points, while the cleaned visualization showed a more balanced distribution. This confirmed that extreme values no longer distorted the analysis. Handling outliers improved the accuracy of the visualizations and business reports. It also ensured that the conclusions drawn from the data were more dependable.

3. What inconsistencies were detected in categorical values and how were they standardized?

The dataset contained inconsistencies such as different spellings, inconsistent letter cases, extra spaces, and duplicate category names. These issues could have resulted in incorrect grouping and inaccurate visualizations. During data cleaning, the values were

standardized by correcting spellings, removing unnecessary spaces, and using consistent naming formats. Tableau and Power BI transformation tools helped apply these changes efficiently. After standardization, similar records were grouped correctly in the reports. This improved the quality and consistency of the overall analysis.

4. **Describe how calculated fields helped in resolving data issues.**

Calculated fields played an important role in preparing clean and meaningful data for analysis. They were used to identify missing values, create outlier flags, and generate corrected values without changing the original dataset. In Tableau and Power BI, calculated fields also simplified repetitive calculations and improved data consistency. They helped automate the cleaning process instead of making manual corrections. This reduced errors and ensured that the reports reflected accurate information. Overall, calculated fields made the dashboard more reliable and easier to maintain.

5. **How did your dashboard demonstrate the effectiveness of the data cleaning process?**

The dashboard clearly showed that the cleaned dataset produced accurate and consistent visualizations without missing values or misleading trends. Charts displayed smooth patterns, correct category names, and meaningful comparisons across different regions and products. Interactive filters worked properly because the data was standardized and complete. The absence of unexpected errors or inconsistent results confirmed that the cleaning process was successful. Users could confidently explore the dashboard and make data-driven decisions. This demonstrated how proper data cleaning improves the quality and usefulness of Business Intelligence reports.

RESULT:

The dataset was successfully cleaned and prepared for accurate analysis using Tableau and Power BI. Issues such as nulls, inconsistencies, and outliers were identified and resolved, improving data quality significantly.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		

